Fiscal Year:	FY 2016	**Task Last Updated:**	FY 02/28/2017
PI Name:	Trivedi, Rohit Ph.D.		
Project Title:	Spatiotemporal Evolution of Three-Dimensional Dendritic Array Structures (SPADES)		
Division Name:	Physical Sciences		
Program/Discipline:	MATERIALS SCIENCE		
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Project Type:	Flight,Ground	**Solicitation / Funding Source:**	NOT AVAILABLE
Start Date:	06/12/2012	**End Date:**	06/11/2016
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA MSFC
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Flight Program:	ISS		
Flight Assignment:	ISS NOTE: Extended to 6/11/2016 per L. Strutzenberg/MSFC and NSSC (Ed., 5/6/15)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Karma, Alain (Northeastern University)		
Grant/Contract No.:	NNX12AK54G		
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Task Description:	Cellular and dendritic microstructures formed during industrial solidification processes (such as casting, welding, and brazing) determine the properties of commercial metallic objects produced in numerous industries, with applications ranging from automobile engine parts to aircraft turbine blades. From a fundamental standpoint, the formation of these structures is controlled by a subtle interplay between microscopic interfacial phenomena, and macroscopic transport of heat and solute, which is strongly influenced by gravity-induced melt convection under terrestrial conditions. Therefore, microgravity directional Bridgman experiments provide a unique means to investigate microstructure development with negligible convection and to test the predictions of theoretical models in a purely diffusive growth regime. While extensive experimental and theoretical work during the past several decades has yielded basic insights into complex solidification patterns, their dynamics in three dimensions (3D) still remains poorly characterized and understood. Fluid convection on Earth leads to large scale inhomogeneities of temperature and alloy concentration in bulk samples. Those inhomogeneities make it difficult to probe the spatially extended pattern dynamics under well-controlled diffusive growth conditions with constant growth velocity, temperature gradient, and alloy composition. For this reason, basic experimental studies of interface dynamics have been restricted to thin samples where fluid flow is eliminated. In addition, computational modeling of 3D patterns has remained challenging due to the several orders of magnitude disparity between capillary and transport scales. Detailed experimental studies in a microgravity environment where (i) convection is reduced sufficiently to obtain growth under diffusive conditions and (ii) the interface evolution can be directly visualized in three dimensions provide an unprecedented opportunity to elucidate fundamental mechanisms of the formation of those hierarchical structures by providing quantitative benchmark data that can be compared to predictions of computational models. The goal of this work is therefore to (i) obtain extensive benchmark experimental data under microgravity conditions over a range of velocity, thermal gradient, and alloy composition in a transparent system by imaging the dynamic evolution of microstructures under directional solidification conditions in the Directional Solidification Insert (DSI) of the DECLIC facility (DEvice for the study of Critical LIquids and Crystallization) installed onboard the International Space Station (ISS), and (ii) use state-of-the-art phase-field simulations to make quantitative predictions of dendritic hierarchical structure formation under diffusive growth conditions that can be directly compared to experimental observations under prescribed growth conditions. We expect the combination of those experimental and computational studies to establish a reliable framework for predicting dendritic microstructures in a wide range of alloy systems of technological relevance.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Many commercial structural alloys, from nickel based superalloys used in turbine blades to light weight aluminum and magnesium alloys used for automotive parts, form dendritic microstructures during the solidification process. These microstructures in turn control the integrity and mechanical properties of cast parts. By bringing together well-controlled and accurately measured three-dimensional solidification experiments with quantitative phase-field simulations, this research should bring significant progress in understanding the origin of side-branching and the nature of the cell to dendrite transition, which have been a long standing issue in solidification for several decades. The deeper understanding of microstructure formation dynamics shall thus accelerate and optimize the design of new technological alloys and their processing conditions that can directly benefit U.S. industries.
Task Progress:	Since no microgravity experiments were carried out during this period of grant, emphasis was placed on the analysis of the experimental images that were obtained from the earlier flight experiments. In addition, efforts were directed towards the development of detailed theoretical models, using the phase field method, to analyze the results and to obtain a reliable and predictive model. Several important ideas were developed from the examination of the experimental results from space. One of the major new result was the discovery of novel oscillating patterns in three-dimensions. For hexagonal arrangement of cells, at certain growth conditions, three pairs of cells were identified that oscillated in a coupled manner with a phase difference of 2pi/3. We developed a model that showed that under some condition there is a gap in spacing where no steady state solutions exist. The oscillating patterns were found to form at the limit of stability of steady-state solutions. Another important and new conclusion was the presence of thermal lag in bulk samples. All theoretical models of initial transient have assumed frozen thermal profile and considered the transient to depend on the build-up of the solute layer only. Our experimental results clearly showed that only solutal considerations give significant departure from the experimental results. A numerical model for thermal lag was used, and an excellent agreement was obtained when both solute and thermal lag were taken into account. Detailed analysis of primary cell/dendrite spacing as a function of velocity was carried out. Two important observations were made. (1) The primary spacing decreased with an increase in velocity, and this decrease in spacing was continuous. In contrast to ground based experiments in thin samples that showed that the primary spacing decreases with increasing velocity, but a sharp increase in primary spacing is observed at the velocity where cell to dendrite transition occurs, no such discontinuity was observed in microgravity experiments. Detailed ground-based studies in thin samples were carried out that clearly showed that the constrained space in thin sample significantly influences the shape of the cells and dendrites and has a significant effect on the array dynamics. The sluggish dynamics often take a very long time to reach steady-state so that metastable structures are present. (2) Careful examination of experimental images showed that the array of cells/dendrites were at a small angle to the growth direction. This very small orientation effect causes the array to translate as new cells are created at one end and some cells are eliminated at the other end. This lateral translation of the array caused the spacing to increase, as observed in experiments in thin samples. NOTE: Project continues with new grant "Spatiotemporal Evolution of Three-Dimensional Dendritic Array Structures (SPADES) -- NNX16AL77G."
Bibliography Type:	Description: (Last Updated: 04/03/2024)
Articles in Peer-reviewed Journals	Tourret D, Debierre JM, Song Y, Mota FL, Bergeon N, Guérin R, Trivedi R, Billia B, Karma A. "Oscillatory cellular patterns in three-dimensional directional solidification." Phys Rev E Stat Nonlin Soft Matter Phys. 2015 Oct;92(4):042401. http://dx.doi.org/10.1103/PhysRevE.92.042401 ; PubMed PMID: 26565251 , Oct-2015

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Articles in Peer-reviewed Journals	Tourret D, Karma A. "Growth competition of columnar dendritic grains: A phase-field study." Acta Materialia. 2015 Jan 1;82:64-83. http://dx.doi.org/10.1016/j.actamat.2014.08.049 , Jan-2015
Articles in Peer-reviewed Journals	Tourret D, Clarke AJ, Imhoff SD, Gibbs PJ, Gibbs JW, Karma A. "Three-dimensional multiscale modeling of dendritic spacing selection during Al-Si directional solidification." JOM. 2015 Aug;67(8):1776-85. http://dx.doi.org/10.1007/s11837-015-1444-2 , Aug-2015